

# NCDB Research Guide

Educational guide | Original web content preserved | External references require internet

A practical learning path from your first research question to a reproducible analysis. You do not need NCDB access or prior Python experience to begin.

## Begin with the foundations

| Order | Lesson                                              | What you will leave with                                             |
|-------|-----------------------------------------------------|----------------------------------------------------------------------|
| 1     | <a href="#">Start Here: What is NCDB?</a>           | A feasible question and a clear description of the population        |
| 2     | <a href="#">Obtaining Access</a>                    | An application preparation checklist and a local approvals record    |
| 3     | <a href="#">The PUF and downloaded files</a>        | A release inventory and a safe storage plan                          |
| 4     | <a href="#">Navigating the Data Dictionary</a>      | Your first documented variable-map entry                             |
| 5     | <a href="#">Coding Systems</a>                      | A plan for checking site, histology, stage, and year-specific coding |
| 6     | <a href="#">Computer, VS Code, and Python Setup</a> | A working environment and a successful synthetic practice run        |

Read these six lessons in order. Each includes a practice task and a readiness check. Use the sidebar to move between lessons or search for a term such as “unknown,” “PUF,” or “clinical stage.”

## Choose your next step

**Still waiting for access?** Complete the foundations and [synthetic walkthrough](#). All practice records are invented.

**Have approved data?** Start with the [import plan](#), [variable map](#), and [cohort definition](#). Keep records in approved institutional storage outside this guide.

**Ready to analyze?** Use the [analysis overview](#) to select a method with your mentor. The later chapters are introductory checklists and the scripts are teaching components, not a validated end-to-end NCDB analysis pipeline.

**Preparing a manuscript?** Review [pitfalls](#), [reporting](#), and [reproducibility and handoff](#).

## How to read this guide

### Official source - verify against your release

Links point to ACS/NCDB or other responsible organizations. Source summaries are original paraphrases, not a replacement for the applicable dictionary, manual, or agreement. Public sources were checked on September 7, 2026; requirements may change. See the [source register](#).

### Educational explanation

Our explanations and fictional examples help you learn the workflow. They do not establish official code meanings or clinical conclusions.

### Project decision - document and review

Eligibility rules, study years, confounders, missing-data methods, model specification, and disclosure review belong to your study protocol. Record who approved each decision and why.

## Keep research data out of the guide

This repository holds code, original teaching material, and synthetic examples only. Never copy patient rows, screenshots of records, model prediction files, credentials, or restricted downloads into it. A private Git repository is also a sharing system. Read [data protection](#) before working with approved records.

## Development status

The six foundation lessons and synthetic workflow are developed for onboarding. Advanced lessons remain explicitly marked teaching outlines. They cover the intended full workflow but still need site-specific examples, expert methodological review, and expanded exercises before serving as a comprehensive analysis curriculum.

# 1. Start Here: What is NCDB?

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## What you will learn

By the end, you should be able to describe the data source, distinguish a case from a population denominator, and decide whether a proposed outcome is observable.

### Official source - verify

The National Cancer Database is a hospital-based registry program of the American College of Surgeons and the American Cancer Society. See the [ACS NCDB overview](#). The [PUF FAQ](#) states that NCDB is not population-based, cannot support population incidence calculations, and that recurrence is not included in the PUF.

## A registry is a structured account of care

**Educational explanation.** A cancer registry translates clinical information into standardized fields. A row represents a cancer case; a column represents an item such as diagnosis year or a treatment characteristic. A registry code is the output of a coding rule, not a complete clinical narrative. “Unknown” may reflect absent documentation rather than a patient’s underlying condition.

The **Participant User File (PUF)** is the research extract you apply to receive. It is not a live connection to hospital charts or the whole registry. The records you analyze have already passed through selection and processing before your study starts. Your analytic cohort adds another layer of selection.

Think of the study population as a sequence:

People with cancer → cases captured by contributing programs  
→ cases included in your PUF release → cases meeting your protocol  
→ cases with information required for a particular analysis

Each arrow can change who is represented. Large sample size does not eliminate selection bias.

## What questions fit?

| Proposed question                                              | Feasibility assessment                                                                   |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------|
| How does recorded treatment use differ across observed groups? | Potentially feasible after verifying treatment definitions and comparable years          |
| Is an exposure associated with overall survival?               | Potentially feasible with valid follow-up, time origin, confounders, and bias assessment |
| What is the national incidence of this cancer?                 | Not an appropriate NCDB PUF question: there is no population denominator                 |

| Proposed question                    | Feasibility assessment                                                                                                |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Does treatment prevent recurrence?   | The required recurrence endpoint is unavailable in the PUF described by the current FAQ                               |
| Did treatment cause longer survival? | Observational comparisons require additional causal assumptions; a regression coefficient alone cannot establish this |

Cancer-specific mortality is also unavailable according to the current [application instructions](#). Do not rename overall survival as cancer-specific survival.

## Turn an idea into a research question

**Project decision.** Write one sentence naming the population, exposure, comparator, outcome, and relevant time period. For example: “Among a prespecified eligible cancer cohort, is recorded treatment A associated with overall survival compared with treatment B?” This is a question template, not a recommended clinical comparison.

Now list what must be observable: eligibility before treatment, treatment definition and timing, outcome timing, baseline confounders, and follow-up. Mark each as verified, uncertain, or unavailable. An uncertain primary outcome is a feasibility problem to resolve before applying or modeling.

Do not select covariates simply because they are available. A confounder can influence both treatment selection and outcome; a mediator occurs along the pathway from treatment to outcome. Adjusting for these has different implications. Bring a timeline or causal diagram to your mentor.

## Practice: a five-minute feasibility note

Write your proposed question, the source population you can actually study, three required concepts, and one limitation that a larger sample would not solve. Find each concept in the release-specific dictionary. Stop at “not verified” when you cannot establish a definition.

## Readiness check

Can you explain why a million hospital cases do not provide a national incidence denominator? Can you name the endpoint your file actually measures? Can you distinguish association from a causal effect? If so, continue to [Obtaining Access](#).

## 2. Obtaining Access

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### What you will learn

Prepare an application with your faculty investigator, locate the institutional contact, and distinguish data access from institutional permission to conduct the study. You can complete the synthetic lessons while waiting.

### Official source - verify before applying

The [ACS PUF portal](#) describes application-based access for investigators associated with CoC-accredited programs. The [June 2026 application instructions](#) specify PI participation, a program support letter, a facility-based analyst, and QPort applicant designation through the program's Primary Contact. Use the live instructions for required fields and deadlines.

### Who does what?

**Educational explanation.** Start by identifying people, not by downloading software.

| Role                                  | Practical preparation task                                               |
|---------------------------------------|--------------------------------------------------------------------------|
| Faculty principal investigator (PI)   | Own the proposal, verify eligibility, and complete required agreements   |
| Student or researcher                 | Draft feasibility notes, learn the dictionary, and document the workflow |
| Cancer registrar / QPort contact      | Help identify the correct program contact and application access route   |
| Analyst or statistician               | Assess data feasibility, design, resources, and analysis requirements    |
| Institutional research/privacy office | Determine local review, storage, access, and retention requirements      |

Being a student at a hospital does not automatically authorize downloading or analyzing its PUF. Work through the PI's approved team and access process.

### Prepare a small application packet

1. Draft the population, exposure, comparator, and outcome in plain language.
2. Verify that the requested organ-site file and diagnosis years can support the question. A familiar disease label is not a substitute for checking file groupings.
3. List required fields and unresolved coding questions in your variable map.

4. Explain the primary analysis, missing-data plan, and principal limitations. Have a mentor assess whether the question is feasible.
5. Assemble personnel information and supporting materials using the current official checklist. Keep biosketches and application records in approved local storage.
6. Ask the QPort contact to confirm the PI's applicant designation; then follow the official portal workflow.
7. Record approval and download notifications, applicable agreements, and the permitted users before obtaining the files.

The [current FAQ](#) restricts data access outside the facility and describes how to update investigators. A collaborator's expertise is not permission to send them data. Resolve team changes with the PI and NCDB.

## **Institutional review is a separate step**

**Project decision.** Obtain your institution's determination about IRB or other research review. Do not copy another paper's "exempt" or "not human subjects research" statement into your protocol. Record the actual determination, date, reference number, and any conditions locally.

Agree where files may reside, who may access them, how backups work, and what happens when a student graduates. Use institutional channels for credentials; do not save passwords in Python, YAML, Markdown, or Git.

## **Your access record**

In approved local storage, maintain a short record with: PI and analyst; authorized team; project purpose; approved file/site; approval reference; storage owner; permitted storage location; training completion; download deadline from the notification; and retention/handoff obligations. The reusable template contains `access_record.md` with blank fields.

## **If access does not work**

Confirm the applicant role with the institutional contact. Follow official password recovery. Do not borrow another person's login. A download archive may use a different password from the portal; follow the instructions supplied with that download.

## **Practice and readiness**

Write a draft email to your mentor identifying the institutional contact, your question, and three unresolved feasibility items. You do not need to send it to complete the lesson. You are ready for the next step when you can identify the PI, access route, local review process, and approved storage owner.

Next: [Understand the PUF and downloaded files.](#)

# 3. Understanding the PUF and Downloaded Files

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## What you will learn

Identify the role of each downloaded file, record the exact release, and explain why a text file needs a matching layout before it can become a trustworthy table.

### Official source - release matters

ACS distributes PUF data as flat text with companion SAS/SPSS scripts; the [PUF resource page](#) links current and legacy materials. The [2024 Getting Started document](#) identifies diagnosis years 2004-2024 for that release. It also notes that item availability depends on diagnosis year. Use the documentation matching the files actually approved for your project.

## Three different dates

**Educational explanation.** A **diagnosis year** belongs to a case. A **PUF release** identifies an edition of the extract. A **download date** records when your team obtained it. They are not interchangeable. A recently downloaded release can contain many earlier diagnosis years, and those years do not guarantee every field was collected throughout.

## Inventory before opening data

| Item                              | Purpose                                    | What to record locally                               |
|-----------------------------------|--------------------------------------------|------------------------------------------------------|
| Compressed/encrypted download     | Delivery container                         | Original filename, release, byte size, download date |
| Extracted text data               | Case records                               | Exact path, checksum, file size; keep unchanged      |
| Layout / import script            | Positions, types, names, and labels        | Release match and script version                     |
| PUF dictionary                    | Meanings, valid codes, availability, notes | Title, revision date, item references                |
| Getting Started and changes notes | Release context and cautions               | Applicable version and issues for your study         |
| Local analysis outputs            | Your transformations                       | Input checksum, code version, run date               |

The project template supplies a blank `file_manifest.csv`. A checksum is a file fingerprint: it helps detect accidental changes, but it does not prove that a file is complete or scientifically valid. Keep manifests describing restricted downloads with the local project.

## Why double-clicking is not importing

A text file may use fixed positions rather than commas. In a fixed-width layout, a field occupies a defined character interval. An incorrect interval can produce plausible-looking but incorrect values without raising an error. Leading zeros can disappear if codes become integers.

The official [2024 Quick Start Data Structure](#) supplies release-specific layout information. This guide intentionally does not reproduce that layout. Follow the [synthetic walkthrough](#) to learn position conversion with invented fields.

For a real parser, reconcile the supplied positions, field lengths, types, encoding, and record length. Validate a small authorized local sample against the supplied import method before processing the whole file. Never use a screenshot or copied patient row to ask for help here.

## Cases, identifiers, and releases

The [2024 Getting Started document](#) describes randomly assigned case identifiers that change between releases and cannot link cases across cancer sites. The [FAQ](#) advises against combining old and new PUF versions.

**Educational explanation.** Do not append a newer release to an older one to “add another year.” Repeated records and updated information can change counts and estimates. Treat each release as a separately documented input. Do not assume one case equals a uniquely linkable person across files.

## Storage plan

Keep real raw files read-only in approved institutional storage outside this repository. Keep processed patient-level files, logs, plots awaiting disclosure review, and model objects there too. A Parquet file is still patient-level data when it contains one row per case. De-identification does not make redistribution appropriate.

**Project decision.** Document a local data path, release, checksums, and parser version. The generated project’s `config.local.yaml` is ignored; its portable `config.yaml` has no real data path. Never place downloaded data under `website/`, where publication tools may copy it.

## Practice and readiness

Without opening any real dataset, fill an inventory for an imaginary release and draw the route from text → validated table → cohort → model. Explain why the diagnosis-year range and variable-year range may differ. Continue to [Navigating the Data Dictionary](#).

# 4. Understanding and Navigating the Data Dictionary

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## What you will learn

Find a field for a clinical concept, check its scope and coding over time, and build a versioned variable map that another researcher can audit.

### Official source - verify each field

Start with the [release-specific PUF dictionary](#) and [legacy documents](#). The 2024 dictionary is revised March 2026. Match your release rather than automatically choosing the newest manual. Registry manuals contain fields that are not necessarily present in a PUF.

## Search by concept, then by exact variable

**Educational explanation.** Suppose your question includes “stage.” That word alone is insufficient: you may need clinical stage before treatment, pathologic stage after surgery, or another defined staging construct. A convenient column name does not settle the scientific question.

1. Write the concept and when it should be measured relative to the exposure.
2. Search the dictionary using the concept and likely synonyms. Read the whole item entry, including notes and related fields.
3. Record the exact raw name and item identifier, where available. Do not rename raw columns in place.
4. Check which cases and diagnosis years the item applies to. Look for changes in definition, coding system, collection requirements, and missing-value conventions.
5. Identify each possible raw value’s meaning using your local documentation. Keep code-like fields as strings during import.
6. Read the matching release changes and linked coding resources. A different manual’s similarly named item may have a different purpose.
7. Write your proposed analytic transformation separately; request mentor or registrar review where the meaning is unresolved.
8. After authorized import, compare actual codes and year-specific frequencies with expectations. Unexpected values should trigger investigation, not automatic coercion.

## A variable map is an analysis contract

The template `variable_map.csv` is a blank metadata table, not patient data. Use one row per source field and coding era when necessary.

| Field group              | What to record                                                                      |
|--------------------------|-------------------------------------------------------------------------------------|
| Concept and raw identity | Concept, exact variable name, item ID, release                                      |
| Source evidence          | Source URL/title, revision, page/section, date checked; summarize in your own words |
| Scope                    | Applicable sites, diagnosis years, definition/coding changes                        |
| Raw representation       | Data type, valid values, distinct unknown/not-applicable codes                      |
| Analytic decision        | New name, transformation, missingness rule, model reference category                |
| Review and validation    | Decision rationale, reviewer, date, checks and unresolved issues                    |

Do not paste entire proprietary code tables into this public guide. Record pointers and the minimal original notes needed to audit your work within permitted use.

## Worked example: fictional codes

Assume an invented field `DEMO_MARKER` uses P for present, A for absent, U for unknown, and N for not applicable. These are **not NCDB codes**.

Your source notes preserve all four meanings. A project decision might produce `marker_present` with 1 for P, 0 for A, and missing for the other two. Keep a companion reason field so U and N remain distinguishable. Mapping both to numeric missing is not permission to forget their different denominators.

```
import pandas as pd
raw = pd.Series(["P", "A", "U", "N"], dtype="string")
assert raw.isin(["P", "A", "U", "N"]).all()
analytic = raw.map({"P": 1, "A": 0}).astype("Int64")
reason = raw.map({"U": "unknown", "N": "not applicable"})
```

If a new value X appears, stop and establish its meaning. Do not convert every unrecognized code to missing silently.

## Build an availability matrix

For each required field, make a row for each diagnosis year or documented era. Distinguish **collected and applicable**, **not collected**, **not applicable**, and **uncertain**. Then inspect observed missingness within the applicable cases. An empty column before collection began is structural unavailability; it is not ordinary sporadic missingness.

**Project decision.** If the exposure is unavailable for part of the proposed period, choose a justified restriction, a documented harmonization, or a revised question. Do not invent an equivalent field or impute an unmeasured historical construct without methodological review.

## Practice and readiness

Complete a map entry for one required concept, including source revision, years, missing codes, a proposed transformation, and an explicit unresolved question. Ask another student to reproduce the decision from your notes. You are ready when they can distinguish official meaning from your analytic choice.

Next: [Coding Systems](#).

# 5. Coding Systems

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## What you will learn

Recognize which reference answers which coding question and avoid treating different years, staging contexts, and tumor schemas as interchangeable.

## A code needs a context

**Educational explanation.** A value has meaning only together with its field, disease context, coding version, and diagnosis year. Codes are labels unless the definition establishes a quantitative scale. For example, a treatment category numbered 3 is not necessarily “three times” category 1.

| System or resource         | Question it helps answer                      | What you must verify                                         |
|----------------------------|-----------------------------------------------|--------------------------------------------------------------|
| ICD-O topography           | Where did the primary tumor arise?            | Applicable site code and version                             |
| ICD-O morphology           | What tumor type and behavior were recorded?   | Histology, behavior, and site compatibility                  |
| AJCC stage                 | What extent of disease is represented?        | Disease schema, edition, clinical/pathologic context, timing |
| STORE / historical FORDS   | How were registry items collected?            | Diagnosis-year manual and whether the item is in your PUF    |
| Collaborative Stage / SSFs | What did an older site-specific factor mean?  | Schema, factor number, year, and definition                  |
| SSDI and grade manuals     | What do site-specific or grade items mean?    | Applicable schema, manual version, timing, and availability  |
| SEER*Rx                    | How is a drug classified for registry coding? | Applicable guidance; do not infer patient regimen detail     |

Official starting points: [SEER ICD-O materials](#), [SEER introduction to site and morphology](#), [ACS registry manuals](#), and the coding-resource links on the [ACS PUF page](#).

## Site is not histology

Topography describes location; histology describes tumor type. Neither alone necessarily identifies the cohort you want. Review the official organ-site groupings and your disease-specific definition before filtering. A file’s name is a starting point for investigation, not a complete eligibility rule.

ICD-O is not interchangeable with ICD-10 diagnosis billing codes. SEER's [ICD-O materials](#) also identify diagnosis-year changes in site-morphology validation resources. Do not apply a current validation table to every historical year without checking applicability.

## Stage is more than a number

Clinical and pathologic assessments use different information at different points in care. A postoperative assessment can be affected by treatment and selection into surgery. Substituting it for baseline disease severity can change the scientific interpretation of a model.

**Project decision.** State the intended stage construct, required timing, precedence rule if multiple fields exist, and approach to discordance. Retain the original fields and a flag identifying the source used. If two staging editions differ, do not assume that identical labels represent identical disease severity. A universal conversion table is not supplied here.

Use the release dictionary and linked AJCC resources to verify the edition and schema. This guide does not reproduce licensed staging tables.

## Changing site-specific fields

A historical numbered site-specific factor and a newer named item may both sound like a biomarker, yet differ in definition, specimen context, collection years, or codes. Check source meaning before combining them. Document a separate transformation for each supported era; otherwise label the era unavailable or restrict the study with a rationale.

## Treatment and grade

Distinguish treatment at the reporting facility from summary treatment where your fields make that distinction. Do not convert a treatment summary into dose, regimen, adherence, or recurrence treatment detail that it does not contain. Likewise, distinguish grade context before combining columns. The correct recode depends on the item and disease, not a generic number range.

## Practice: the same code in two eras

In an invented dataset, era A uses 1 = marker present; era B uses 1 = test not performed. A naïve merge would combine incompatible categories. Write two mapping rules and a reason for every missing value. Then explain why looking only at the distribution of 1 cannot detect this mistake.

## Readiness check

For one planned field, identify its coding resource, version, disease scope, years, and timing. If any is unknown, put the uncertainty in your variable map before analysis. Next: [Computer Setup](#).

# 6. Computer, VS Code, and Python Setup

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## What you will learn

Create an isolated Python environment, select it in VS Code, render the website, and run a practice analysis without real data.

## Know the tools

**Educational explanation.** Python runs analysis code. VS Code edits files and provides a terminal. A terminal accepts commands; it is not the Python prompt marked `>>>`. Git records source changes. Quarto converts `.qmd` lessons into a searchable website. Installing the Python package named `quarto` does not replace the Quarto CLI.

Use Python **3.11** for the tested environment. The dependency snapshot records an exact tested package set; other Python versions and operating systems require their own validation. Download installers from [Python](#), [VS Code](#), [Git](#), and [Quarto](#), following institutional software policy.

## Open the right folder

In VS Code, choose **File → Open Folder** and select the guide folder containing `requirements.txt`, `create_ncdb_project.py`, and `website`. Open **Terminal → New Terminal**. Commands below assume that folder is your working directory. Do not type the Markdown backticks or the prompt symbol.

## Create and install the environment

On macOS/Linux:

```
python3.11 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
```

On Windows PowerShell:

```
py -3.11 -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install -r requirements.txt
```

If activation is blocked by institutional policy, call the interpreter directly rather than changing that policy:

```
.\.venv\Scripts\python.exe -m pip install -r requirements.txt
```

A virtual environment holds this project's packages separately. Recreate it after moving the project; do not copy it or commit it. See [Python's venv documentation](#).

## Select the interpreter in VS Code

Install Microsoft's Python extension. Open the Command Palette and run **Python: Select Interpreter**; select the Python inside this folder's .venv. Open a fresh terminal. The [VS Code environment guide](#) documents interpreter selection and environment discovery.

```
python -c "import sys; print(sys.executable); print(sys.version)"
python -m pip check
python tools/check_environment.py
```

The executable should end in .venv/bin/python or .venv\Scripts\python.exe. pip check should report no broken requirements. If an import fails, compare the selected interpreter with the one printed above before reinstalling anything.

## Build and search the guide

```
quarto --version
quarto render website
quarto preview website
```

A successful render produces website/\_site/index.html and search.json. Preview starts a local server; open the address printed in the terminal and try searching "variable map." Stop preview with Ctrl+C. The website displays code examples without executing analysis on render; executable practice is tested separately.

If this machine's Quarto was installed locally during development, use the absolute CLI path recorded in AUDIT.md, or install the CLI normally from the official download page. Restart the terminal after installation if quarto is not found.

## Run the synthetic practice project

```
python create_ncdb_project.py --name "Practice study" --site "Synthetic" --outcome "demo_event" --exposure "demo_group"
python projects/practice_study/scripts/run_synthetic.py
python -m unittest discover -s tests -v
```

The generator creates a protected local project. The demo creates invented records in memory and writes only synthetic aggregate outputs into its ignored results/ folder. It does not read NCDB files. See the [walkthrough](#) for interpretation.

## A few Python ideas

A variable names an object. A function packages a reusable operation. A pandas DataFrame is a table; a Series is a column. df["age"] selects a column; df.loc[mask] selects rows where a Boolean condition is true. Indentation defines blocks. A traceback ends with the immediate error, but the earlier lines show how execution reached it.

## Storage and readiness

**Project decision.** Use only institution-approved storage for real analysis and review [data protection](#). Do not point practice scripts at patient records. You are ready when the environment check, website render, and synthetic run succeed and you can locate their outputs.

# Synthetic Walkthrough

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## Purpose

Practice a reproducible sequence before handling approved records. Every field beginning `demo_` is invented; the distributions and model coefficients have no NCDB or clinical interpretation.

## Run it

From the guide root with the Python environment active:

```
python create_ncdb_project.py --name "Practice study" --site "Synthetic" --outcome "demo_event" --exposure "demo_group"
python projects/practice_study/scripts/run_synthetic.py
```

The generator refuses to overwrite an existing project. To repeat the practice analysis, run the second command again. Outputs are overwritten only in that project's synthetic results folder.

## What happens

1. An in-memory fixed-width example uses invented ID, age, and marker fields. The first ID remains 001, demonstrating why codes should initially be strings.
2. A seeded random generator creates 600 fictional records with age, group, a separate binary outcome, and censored follow-up.
3. A fictional age restriction selects an adult practice cohort and records exclusions. This rule is not a recommendation for your study.
4. The script writes missingness summaries, grouped counts, logistic estimates, a KM plot, Cox estimates, and a small run summary.
5. Propensity scores are estimated to exercise the function, but no matching or causal effect is claimed. Individual scores and individual records are not exported.

## Learn fixed-width positions

Imagine 00145P: positions 1-3 are an invented identifier, 4-5 are age, and 6 is a fictional marker. Python's `colspecs` use zero-based, end-exclusive positions: [(0, 3), (3, 5), (5, 6)]. The endpoint is not included. This example is **not** an NCDB layout. See [pandas read\\_fwf](#).

## Inspect the outputs

Open `projects/practice_study/results/synthetic/summary.json` and `cohort_flow.csv` locally. Confirm that retained plus excluded equals 600. Check that Table 1 counts sum to the final cohort. The binary outcome used for logistic regression differs from the follow-up event used by survival models; they answer different demonstration questions.

The KM curve displays censoring-based estimates for invented groups. The logistic and Cox confidence intervals reflect fitted sampling models, not real clinical evidence. These estimates do not validate assumptions or authorize the same specification on NCDB data.

## **Extend carefully**

Change the synthetic seed and observe variation. Add a missing predictor and confirm that the model helper refuses it. Explain why that refusal requires a missing-data decision rather than an automatic `.dropna()`. Run tests again after changing utilities.

Next: use the [analysis overview](#) to plan a real study with your mentor.

# Importing NCDB Data

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## Goal

Convert the raw local NCDB file into an analysis-ready local format.

## Recommended pipeline

```
Raw NCDB file
↓
Read using supplied file structure
↓
Assign names and data types
↓
Validate row/column counts
↓
Audit missingness
↓
Save processed local copy
```

Parquet is convenient for repeated Python analyses because it preserves data types and loads quickly.

```
import pandas as pd

df = pd.read_parquet("data/processed/ncdb_master.parquet")
```

The exact parser should be adapted to the locally supplied NCDB file format and documentation.

# Building the Cohort

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## Define the cohort before running models

Write inclusion and exclusion criteria in plain language before coding them.

Example:

Initial disease-site cohort

- diagnosis years
- age restriction
- histology restriction
- known exposure
- known outcome
- valid staging
- analytic cohort

The script should record the number of patients remaining after every step.

# Cleaning and Recoding

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## Key principles

- Keep raw variables unchanged
- Create new analytic variables
- Document every recode
- Distinguish missing, unknown, and not applicable
- Avoid duplicate representations of the same construct
- Check coding changes by diagnosis year

Example recodes must follow the release-specific variable map. Keep an age top-code or unknown code from becoming an ordinary continuous age accidentally. Use the fictional mapping in the [dictionary lesson](#) to practice without importing real records.

# Missingness and Data Quality

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## Audit in layers

Start with raw-code frequencies and valid-value checks, then distinguish unknown, not applicable, not collected, suppressed and ordinary blank values where your dictionary supports those meanings. Python's `isna()` sees software missing values; it does not know registry conventions.

Inspect availability and missingness by diagnosis year, disease scope and exposure group. Check ranges, timing consistency and unexpected categories. Keep an explicit reason field when multiple raw codes become analytic missing. Exact duplicate rows are a signal to investigate identity and import behavior, not automatic permission to delete.

## Document the analysis denominator

Write down input N, exclusions and reasons, final N and events for each model. Complete-case analysis can change the population and introduce bias. Imputation cannot manufacture a historically uncollected construct without substantial assumptions. Choose and review the missing-data strategy before fitting; model helpers refuse missing input so this decision cannot be hidden by default deletion.

The [dictionary lesson](#) provides the map and availability workflow.

# Analysis Overview

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## Select a method from the question

The templates are teaching components. Read `ANALYSIS_GUIDE.md` in your generated project for required variables, assumptions, NCDB pitfalls, code behavior and interpretation for every script. The synthetic demo tests mechanics; it does not validate a clinical analysis.

| Task                                 | Begin with                         | Decision that must precede code                           |
|--------------------------------------|------------------------------------|-----------------------------------------------------------|
| Understand the cohort                | Missingness, descriptive summaries | Denominators, coding validity and eligibility             |
| Describe group characteristics       | Table 1                            | Category definitions and missingness reporting            |
| Model a binary endpoint              | Logistic regression                | Outcome window, confounders, encoding and functional form |
| Describe time to event               | Kaplan-Meier                       | Time origin, event/censor coding and groups               |
| Adjust a time-to-event comparison    | Cox regression                     | PH, timing, confounders and facility correlation          |
| Balance measured baseline covariates | Propensity methods                 | Target estimand, overlap, design and inference            |
| Challenge assumptions                | Sensitivity analyses               | Named assumption and justified alternate rule             |
| Develop prediction                   | Machine learning                   | Prediction target, time horizon and validation strategy   |

## Use an explicit sequence

Define the question and protocol → verify variables → import and audit → build the cohort → recode → inspect missingness and Table 1 → implement the approved primary analysis → assess diagnostics → run prespecified sensitivities → review outputs and reporting.

Real-data model helpers do not select covariates, drop missing rows, tune definitions or publish results. Facility clustering, missing-data alternatives, causal estimands and treatment timing require additional study-specific implementation and expert review.

# Descriptive Analysis

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## Start here before regression

Always inspect:

- cohort size
- missingness
- distributions
- implausible values
- category frequencies
- outcome prevalence
- exposure prevalence

Do not fit the final model until these checks are complete.

# Regression

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## Logistic regression

Use for binary outcomes.

Report:

- adjusted odds ratio
- 95% confidence interval
- p-value
- reference category
- sample size
- model specification

## Linear regression

Use when assumptions are appropriate for continuous outcomes.

Always check the distribution and residual behavior.

# Survival Analysis

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## Common tools

- Kaplan-Meier curves
- log-rank test
- Cox proportional hazards regression

## Always check

- time origin
- event definition
- censoring
- follow-up availability
- proportional hazards assumption

# Propensity Methods

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## Goal

Improve measured covariate balance between exposure groups.

Possible approaches:

- matching
- inverse probability weighting
- overlap weighting

## Required diagnostics

- standardized mean differences
- pre/post balance
- retained sample size
- overlap/common support

# Sensitivity Analyses

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## Challenge a named assumption

A sensitivity analysis asks whether a conclusion depends on a defensible alternative assumption. Examples include an alternate coding-era restriction, a different exposure definition, a justified missing-data approach, or an analysis addressing treatment timing.

For each analysis record the assumption, changed rule, target population/estimand, expected change in sample, and comparison to the primary result. Compare effect sizes, confidence intervals and cohort composition. A change in statistical significance alone is not a robustness criterion.

Prespecify the main alternatives and label exploratory additions. Do not cycle through exclusions to obtain a preferred p-value. Unmeasured confounding, immortal time and structural unavailability require design-specific expertise; a generic loop over models will not solve them.

# Machine Learning

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## Secondary analyses

Machine-learning methods can supplement, but should not automatically replace, a clinically interpretable primary model.

Possible uses:

- prediction
- nonlinear relationships
- feature importance
- SHAP analysis

Always separate predictive importance from causal interpretation.

# Figures and Tables

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## Standard outputs

- Cohort flow diagram
- Table 1
- Regression results
- Kaplan-Meier curves
- Forest plots
- Sensitivity analyses

All exported outputs should be reproducible from scripts.

# Manuscript Reporting

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## Methods checklist

Describe:

- data source
- study design
- diagnosis years
- inclusion/exclusion criteria
- exposure
- outcome
- covariates
- missing-data strategy
- statistical methods
- sensitivity analyses

Avoid wording that implies population-based incidence unless appropriate for the data source.

# Common NCDB Pitfalls

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## Common errors

- treating numeric codes as continuous values
- mixing clinical and pathologic stage without a prespecified rule
- ignoring diagnosis-year coding changes
- failing to distinguish unknown from absent
- using overlapping predictors
- overadjusting for mediators
- interpreting association as causation
- starting analysis before cohort definition
- ignoring missing-data patterns
- failing to validate derived variables
- sharing restricted patient-level data

# Reproducibility and Project Handoff

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## Reconstruct the work

Record the PUF release and local input checksums, dictionary revisions, protocol version, source-code commit, Python/package versions, Quarto version and run command. Keep patient-level inputs and outputs in approved storage. The public code should be understandable without copying those records.

Separate immutable inputs, intermediate transformations and reviewed aggregate results. Preserve a cohort flow and a record of changes to the protocol. A fixed random seed makes a computational step repeatable; it does not address biased study design.

## Hand off responsibility

Give the next authorized researcher the question, access requirements, approved storage location, environment instructions, variable map, cohort rules, analysis plan, diagnostics, disclosure review record and known limitations through approved channels. Test recreation using the synthetic demo. Re-run the real study only in the authorized environment.

Have the PI confirm access changes when a researcher leaves. Do not transfer credentials, copy data to personal storage, or assume access continues after graduation.

# Data Protection and Safe Sharing

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## Repository boundary

This guide contains original text, source code, blank metadata templates and invented examples. Keep actual NCDB records in institution-approved storage outside the guide. Patient-level derived tables, predictions, matched cohorts, imputation files, model objects, notebooks, logs and screenshots can be sensitive too.

The `.gitignore` blocks common data formats, archives, credentials, environments, website output and generated projects. Only named blank CSV metadata templates are exempted. Do not turn those templates into patient tables. `.gitignore` does not remove already-tracked files and can be bypassed with `force-add`; it is one guard, not a security boundary.

## Before a commit

After staging, run `python tools/check_repository.py` from the guide root. It checks the Git index, including staged content, and blocks common risky formats, local paths, oversized files and some credential patterns. Review `git diff --cached` and `git diff --cached --stat` locally. Never paste a diff containing data into a public review tool.

For a folder without Git metadata, `python tools/check_repository.py --source-tree` checks source files while excluding local/generated directories. It is not a full disk audit. The tool cannot recognize every secret, patient record hidden in text, or disclosure through aggregate results.

To enable the included local pre-commit hook after initializing Git:

```
git config core.hooksPath .githooks
```

Run commits from an environment where `python` resolves to your project interpreter. Hooks can be bypassed; the included CI guard provides a second check, not authorization to upload records.

## Publication needs a separate review

### Official source - verify your agreement

The current [PUF FAQ](#) discusses restrictions on reporting small cells and the need to prevent reconstruction. Apply your executed agreement and institutional requirements; an example threshold in code is not the governing rule.

**Educational explanation.** If a total is 100 and one visible category is 97, hiding the other category does not hide its count of 3. Percentages, margins, complementary categories, repeated tables and KM risk tables can reveal counts. The suppression helper only formats one count and requires an explicit threshold. It does not perform complementary suppression or certify a table.

**Project decision.** Name a reviewer, applicable rule, approved output destination, and review date. Publish only cleared aggregate artifacts. Do not publish the entire analysis directory or its generated project.

## **If data were staged or shared**

Stop the commit/upload. Keep incident details in approved channels and notify the responsible PI/privacy or security team. Removing a file from the working tree does not remove history, remote copies, logs or backups. Follow institutional incident procedures; do not attempt an undocumented history rewrite or claim that a deleted file has been recalled.

# Official Sources and Verification Register

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Public links reviewed **September 7, 2026**. These are references, not copied manuals. Link availability and release content can change; recheck before applying, importing, recoding or reporting. This educational project is not an official ACS publication.

| Resource                                        | Use                                                       | Version or review note                         |
|-------------------------------------------------|-----------------------------------------------------------|------------------------------------------------|
| <a href="#">ACS NCDB overview</a>               | Data-source description                                   | Live overview                                  |
| <a href="#">ACS PUF resources</a>               | Current and legacy releases, agreements, coding resources | Landing page linked 2024 resources at review   |
| <a href="#">Application instructions</a>        | Roles, access route, application elements                 | June 2026                                      |
| <a href="#">PUF FAQ</a>                         | Access, scope, release and reporting cautions             | May 2026                                       |
| <a href="#">Getting Started with 2024 PUF</a>   | Release and case context                                  | March 2026                                     |
| <a href="#">2024 Data Dictionary</a>            | Item definitions and availability                         | Revised March 2026; check your own release     |
| <a href="#">2024 Quick Start Data Structure</a> | Layout reference                                          | Release-specific; no positions reproduced here |
| <a href="#">2024 PUF Changes</a>                | Release change review                                     | Review before porting prior code               |
| <a href="#">ACS registry manuals</a>            | STORE and historical references                           | Choose diagnosis-year manual                   |
| <a href="#">SEER ICD-O materials</a>            | Coding updates and validation resources                   | Match diagnosis year/version                   |
| <a href="#">SEER coding introduction</a>        | Site and morphology explanation                           | Educational reference                          |
| <a href="#">Python venv</a>                     | Environment isolation                                     | Select documentation for installed Python      |
| <a href="#">VS Code environments</a>            | Interpreter selection                                     | Interface may evolve                           |
| <a href="#">Quarto download</a>                 | CLI installation                                          | Validated locally using 1.9.38                 |

| Resource                                        | Use                        | Version or review note                   |
|-------------------------------------------------|----------------------------|------------------------------------------|
| <a href="#">pandas read_fwf</a>                 | Fixed-width parser API     | Explicit positions required for practice |
| <a href="#">statsmodels Logit</a>               | Logistic implementation    | Consult installed-version behavior       |
| <a href="#">lifelines survival regression</a>   | Cox assumptions and API    | Consult installed-version behavior       |
| <a href="#">scikit-learn LogisticRegression</a> | Propensity-score component | Regularized classifier, not matching     |

## Maintaining evidence

For an official claim, record organization, document title, version/revision, section, URL, and date verified. For an educational example, mark it fictional and explain its purpose. For a project decision, record rationale, reviewer, and protocol version. Unresolved discrepancies belong in the variable map and should be resolved with the responsible expert before analysis.

No official manual, complete coding table, licensed stage definition, or patient record is bundled with this guide.

# Troubleshooting

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## Common programming problems

### File will not load

Check:

- path
- delimiter
- encoding
- expected number of columns
- supplied import/layout script

### Regression fails

Check:

- missing values
- outcome coding
- perfect separation
- duplicate predictors
- sparse categories

### Unexpected sample-size loss

Print the cohort size after every filtering step.